

The background of the slide is a light gray gradient. It is decorated with several realistic water droplets of various sizes. Some droplets are at the top left, some are at the bottom right, and others are scattered in the center. Each droplet has a highlight and a shadow, giving it a three-dimensional appearance.

# LOGIC GATES

STANDARD SYMBOLS

# LOGIC GATES

- THERE ARE 4 BASIC LOGIC GATES
  - AND
  - OR
  - NOT
  - X-OR (EXCLUSIVE OR)
- ADDITIONALLY THERE ARE 3 DERIVED GATES
  - NAND
  - NOR
  - X-NOR

# AND GATE

| x | y | x AND y |
|---|---|---------|
| 0 | 0 | 0       |
| 0 | 1 | 0       |
| 1 | 0 | 0       |
| 1 | 1 | 1       |



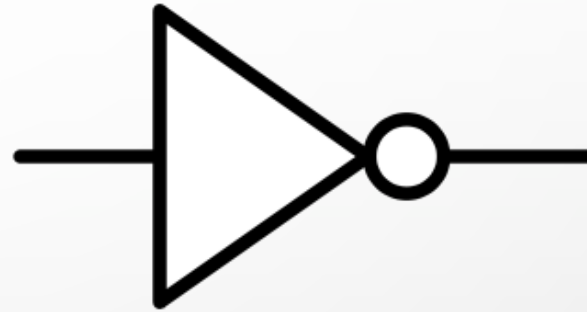
# OR GATE

| x | y | x OR y |
|---|---|--------|
| 0 | 0 | 0      |
| 0 | 1 | 1      |
| 1 | 0 | 1      |
| 1 | 1 | 1      |



# NOT GATE

| x | NOT x |
|---|-------|
| 0 | 1     |
| 1 | 0     |



# X-OR GATE

| a | b | a X-OR b |
|---|---|----------|
| 0 | 0 | 0        |
| 0 | 1 | 1        |
| 1 | 0 | 1        |
| 1 | 1 | 0        |



# NAND GATE

| x | y | x NAND y |
|---|---|----------|
| 0 | 0 | 1        |
| 0 | 1 | 1        |
| 1 | 0 | 1        |
| 1 | 1 | 0        |



# NOR GATE

| x | y | x NOR y |
|---|---|---------|
| 0 | 0 | 1       |
| 0 | 1 | 0       |
| 1 | 0 | 0       |
| 1 | 1 | 0       |



# X-NOR GATE

| a | b | a X-NOR b |
|---|---|-----------|
| 0 | 0 | 1         |
| 0 | 1 | 0         |
| 1 | 0 | 0         |
| 1 | 1 | 1         |

